

Chapter 1

Triangles

1.1 Defintion

define as the union of three line segments. the three segments must be unique.

define inside and outside of the triangle based on if a point x is on the same side of each comprising segment or not

Axiom 1 (SSS Congruency). *Given two triangles $\triangle ABC$ and $\triangle XYZ$, if all sides of one are congruent to a respective side of the other triangle, the triangles themselves are considered congruent:*

$$\triangle ABC \cong \triangle XYZ.$$

For congruent triangles, corresponding parts are congruent.